Theory-Based Augmented User Modeling with Grounded Transformers

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Abstract. The user model serves as a cornerstone of Intelligent Tutoring Systems. In many instances, this modeling relies on Knowledge Tracing approaches, which represent the learner's state of mastery across specific skills or knowledge components. However, these approaches are not anchored to theoretical frameworks derived from learning theories, which limits their utility when it comes to interpreting user profiles or explaining the evolution of learning trajectories. To overcome these limitations, we propose a new approach to user modeling based on Grounded Transformers, a class of deep learning models whose internal representations are anchored to interpretable models based on cognitive or pedagogical principles. The resulting user models support instructional decisions rooted in learning theories, and we demonstrate how this grounding enables personalised, context-aware instruction in Intelligent Tutoring Systems.

Keywords: deep learning · theory-based interpretability · grounded transformers · user modeling · learning theories · intelligent tutoring systems

1 Introduction

1.1 Problem Statement

The student model serves as a cornerstone of Intelligent Tutoring Systems. In many instances, this modeling relies on Knowledge Tracing (KT) [3] approaches, which represent the learner's state of mastery across specific skills or knowledge components. However, these traditional approaches are subject to significant limitations since they lack pedagogical interpretability, as they are not anchored to theoretical frameworks derived from cognitive or educational principles.

To overcome these limitations, Grounded Transformer have been recently proposed as a class of deep learning models able to provide information about the evolution of the user according to parameters derived from theoretical principles. In [8] they introduce gTransformer, and show how through *representational grounding* it is able to anchor its latent representations to pedagogically meaningful concepts based on Bayesian Knowledge Tracing (BKT).

In this work, we show how grounded transformers can be leveraged to create augmented student models to improve instructional decisions in Inteligent Tutoring Systems. To this end we propose gBKT, a traditional BKT model augmented with output from a gTransformer model, to enrich the student model. We evaluate gBKT, show its advantages compared with a traditional BKT model and how it can be used to create student models able to support better instrucional decisions.

1.2 Research Contributions

This work makes three primary contributions:

- *Augmented Theory-Based Models*: We introduce a methodology for anchoring deep learning models to Bayesian Knowledge Tracing (BKT) logic, enabling the generation of pedagogically grounded parameters (p_{L_0}, p_{T}) that capture individualised student characteristics while maintaining high predictive accuracy.
- *Augmented Student Model*: We propose an augmented user model that integrates these grounded parameters to identify four distinct *learning situations—foundational*, *emerging*, *consolidating*, and *advancing*—based on each student's prior knowledge and learning rate, providing a higher-resolution diagnostic profile than population-level models.
- *Adapted Instructional Strategies*: We demonstrate how this augmented user model enables situation-based instruction, where the ITS tutorial model can dynamically route students to differentiated instructional pathways tailored to their specific learning situation.

1.3 Research Questions

This work is guided by following research questions that together examine how Grounded Transformers [8] can be leveraged to improve theory-based augmented user modeling:

- RQ1: How can we obtain augmented student models with information derived from existent learning theories?
- RQ2: How these augmented student models can be used to improve Intelligent Tutoring Systems?

2 Background

2.1 Intelligent Tutoring Systems and User Models

Intelligent Tutoring Systems are typically structured around three interacting components:

- *domain model*: This component encodes the knowledge to be taught, typically structured as a set of Knowledge Components (KCs) or skills. It provides the semantic framework through which student performance is assessed and instructional content is organised.
- *user model*: The user model tracks the learner's evolving state by maintaining a representation of their mastery across the domain model. In gTransformer, this is augmented with longitudinal parameters that capture the student's unique learning patterns beyond simple correctness.
- *tutorial model*: This component mediates instructional decisions by selecting the next activity, intervention, or level of support based on the current state of the user model. The goal is to provide a personalised learning experience that optimises the student's trajectory.

2.2 Learning Theories and Bayesian Knowledge Tracing

We select BKT [3] as our reference theoretical framework. BKT models the learning process as a Hidden Markov process governed by four interpretable parameters: Initial Mastery ($P(\mathrm{L}_0)$), the probability that a student knows a concept prior to practice; Learn Rate ($P(\mathrm{T})$), the probability of transitioning from the unlearned to the learned state; Guess ($P(\mathrm{G})$), the probability of a correct response despite lack of mastery; and Slip ($P(\mathrm{S})$), the probability of an incorrect response despite mastery. These parameters enable a recursive, causally transparent estimation of student mastery that is directly interpretable by educators.

Standard BKT suffers from a significant limitation: it operates at the population level, estimating a single parameter set for all students on a given skill [12]. gTransformer addresses this by leveraging Transformer attention mechanisms to infer student-specific parameters from individual interaction histories.

2.3 Grounded Transformers

3 Theory-Based Augmented User Models

We formalise the proposed ITS architecture by integrating traditional components with our grounded modelling engine. The *domain model* defines the structured knowledge components to be mastered. The *user model* tracks student mastery across these components, and is augmented with student-specific parameters ($p_{\mathrm{L}_0}, p_{\mathrm{T}}$) provided by *gTransformer*.

The *gBKT* component performs Bayesian updating of student mastery L_t, where the initial belief L_0 and the learning transition T are dynamically set by gTransformer's grounded outputs. These parameters encapsulate the student's longitudinal learning context, allowing the user model to differentiate between learners who might show similar correctness patterns but possess different underlying learning rates or prior knowledge. The output of this augmented user model is used by: 1) the tutorial model to select instructional strategies, and 2) a diagnostic roster module that visualises student evolution in a 3D $(p_{\mathrm{L}_0}, p_{\mathrm{T}}, t)$ space.

4 Methodology

To address RQ1, we evaluate *gBKT* by comparing its predictive performance against traditional population-level BKT. Unlike standard BKT which uses constant parameters for all students on a given skill, gBKT leverages gTransformer to infer context-aware parameters from historical interaction sequences. gTransformer uses a Transformer-based encoder-decoder architecture where latent representations z_t are projected onto BKT semantic axes through representational grounding. The model is trained on benchmark datasets such as AS2009, with the grounded parameters serving both to constrain predictions to pedagogical logic and to augment the student profile.

To address RQ2, we define four *learning situations* based on the estimated $(p_{\mathrm{L}_0}, p_{\mathrm{T}})$: (1) *foundational learners* (low L_0, low T), (2) *emerging learners* (low L_0, high T), (3) *consolidating learners* (high L_0, low T), and (4) *advancing learners* (high L_0, high T). These situations allow the tutorial model to adapt strategies: foundational learners might receive targeted scaffolding, while advancing learners receive accelerated content. Since parameters are dynamic, students can transition between learning situations as more evidence is gathered. To illustrate this, we generate 3D roster plots showing the longitudinal trajectory of representative students across these learning situations.

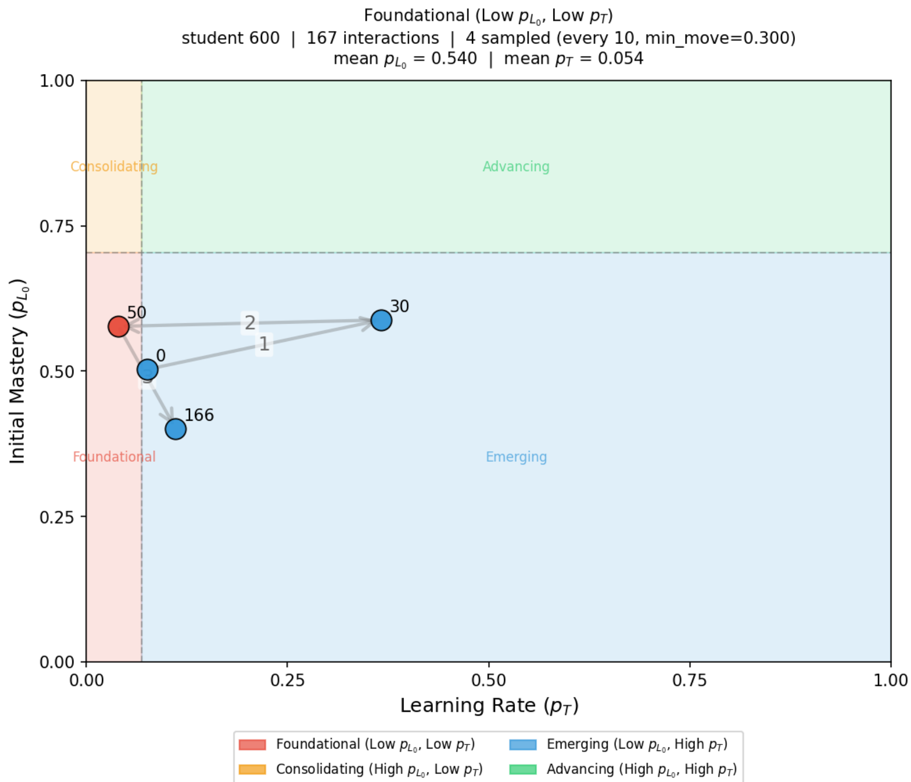
5 Results and Discussion

Comparison of predictive performance (Table 1) addresses RQ1, showing that gBKT significantly outperforms standard BKT, with an average AUC gain of 19.9% ($\mathcal{I}_{\mathrm{gain}}$). This validates that context-aware parameters derived from grounded Transformers accurately capture the nuances of individual learning journeys while remaining within the interpretable framework of BKT.

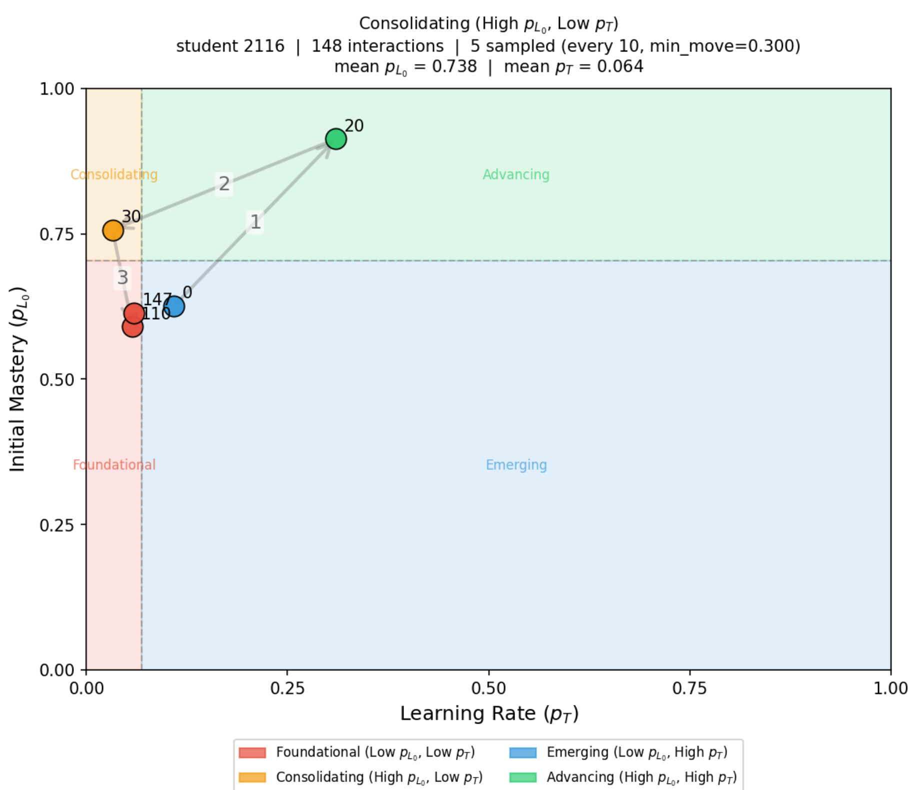
Figure 1 addresses RQ2, showing trajectory plots for a representative student from each of the four learning situations. Each panel maps consecutive sampled interactions in (p_T, p_{L_0}) space, with numbered points and directional arrows tracing the temporal evolution of the student's parameter profile across the full interaction sequence. Coloured background regions delimit the four learning situations; a point may cross boundaries, indicating a genuine shift in the student's learning situation over time. By reading these trajectories, educators gain theory-grounded insights into whether a student is disengaging (decreasing p_{T}) or gaining confidence (increasing p_{L_0}). This dynamic view grounds deep learning state into pedagogically recognisable constructs, enabling more mindful instructional support and satisfying the accountability requirements for high-risk educational AI.

The parameter orbits of representative students across the four learning situations are characterised in Figures 2 and 3. A key property of gTransformer's augmented user model is that the learning situation label assigned to each student is not a static category frozen at enrolment: it is a summary of the student's longitudinal tendency, continuously updated as new interactions arrive. The covariance ellipses in Figure 2 make this explicit—each ellipse visualises the full spread of $(p_{\mathrm{L}_0}, p_{\mathrm{T}})$ values that a student's profile traverses across all their interactions, rather than a single point estimate.

For *foundational learners*, the ellipse is compact and centred at low mastery and low learning rate, indicating limited contextual variation: across skills and interactions, the model consistently reads this student as struggling to progress. This persistence is itself instructionally informative—it is not that no learning is occurring, but that the rate is systematically insufficient given the current instructional conditions. The tutorial model should not wait for cumulative failure before acting; the ellipse signals a structural mismatch between task demands and student preparation that calls for immediate scaffolding, worked examples,



(a) *Foundational* (low p_{L_0}, low p_T). The trajectory remains confined to the lower-left quadrant throughout the interaction sequence. The small number of sampled points and their limited spread indicate that the student's parameters change very little: neither mastery nor learning rate improves substantially, signalling a structural mismatch between task demands and current preparation that calls for immediate scaffolding.



(b) *Consolidating* (high p_{L_0}, low p_T). The trajectory is compact and anchored in the and reduced problem complexity. Crucially, this diagnosis can be revised: if the student responds to the intervention and p_{L_0} or p_{T} begins to rise, the orbit will shift and the learning situation label will update accordingly.

For *consolidating learners*, the ellipse is compact and located at high mastery and low learning rate. The small spread indicates that the student has reached a stable region of the parameter space with little contextual variation: prior knowledge is high but the per-interaction learning rate has converged to near zero. This pattern is not a failure—it reflects a student who has consolidated existing knowledge—but it does signal stagnation. The tutorial model should interpret this as a cue to introduce novel, cross-domain challenge or to advance the student to a higher difficulty tier, as continued practice at the current level is unlikely to produce further gains. If challenge is appropriately increased and p_{T} recovers, the ellipse will redistribute and the tutorial model will update its routing.

For *emerging learners*, the ellipse is the most spatially extended of all four situations, oriented along both the mastery and learning-rate axes. This broad orbit reflects a student whose parameters vary substantially across skills and interaction contexts: at some skills the student is already proficient; at others, they are still building foundations, but with a high and increasing learning rate. This heterogeneity is precisely what a static label would erase. The tutorial model should exploit it by differentiating at the skill level—providing scaffolding on weaker knowledge components while allowing acceleration on stronger ones—rather than applying a uniform strategy to the whole student.

For *advancing learners*, the ellipse is moderately extended and centred at high mastery with a moderate learning rate. The spread reflects a student who engages responsively across varied skill demands while maintaining consistently high prior knowledge. The tutorial model can safely route this student to accelerated content and reduced practice redundancy; the ellipse shape confirms that the high-mastery profile is robust across contexts and not an artefact of easy items.

Figure 3 shows return maps (lag plots) for the same four representatives. Each panel plots the value of a parameter at interaction $t + 1$ against its value at interaction t, separately for p_{L_0} (left sub-panel) and p_{T} (right sub-panel), with points coloured from dark (early interactions) to bright (late interactions) using a sequential palette. The diagonal represents a fixed point: points on the diagonal indicate no change between consecutive interactions. Deviations above the diagonal signal an increasing parameter; deviations below signal a decrease. Unlike population-level BKT, which produces a single pair of constant parameter values for all students on a given skill, each return map encodes the unique temporal trajectory of an individual student—a view that is both dynamic and context-aware, and one that evolves measurably as the student progresses.

For *foundational learners*, the p_{L_0} return map reveals a broad scatter around the diagonal with a systematic downward drift from early interactions (mean 0.61) to late interactions (mean 0.48): the model progressively revises its mastery estimate downward as evidence of insufficient learning accumulates. The

p_{T} panel is tightly concentrated near zero throughout, exposing a stable low-rate attractor that a snapshot profile would reduce to a single low number. The trajectory, however, shows that the attractor consolidates over time—the orbit is converging, not oscillating randomly. This temporal structure informs instructional timing: an intervention delivered before the orbit locks into the low attractor is far more likely to dislodge the student from this trajectory than one applied after convergence. The tutorial model, monitoring this return map in real time, can act early and revise the learning-situation assignment as soon as p_{T} or p_{L_0} shows a sustained upward deviation.

For *consolidating learners*, the p_{L_0} return map undergoes a visible temporal transformation: early points span the mid range, while late points consolidate near the diagonal at high mastery (0.80), forming a tight cluster. Simultaneously, p_{T} declines from 0.077 to 0.050, converging toward a low-rate fixed point. The orbit has not always been in this state—the student arrived at the plateau through accumulated practice—and the return map preserves that history rather than collapsing it into a single label. A static model would see only high mastery and low learning rate; the dynamic view reveals that this student is transitioning from active learning to consolidation. The tutorial model can exploit this information to introduce transferable challenge or cross-domain tasks that reactivate the learning trajectory before the orbit fully stabilises, rather than treating the plateau as an irreversible state.

For *emerging learners*, both p_{L_0} $(0.52 \rightarrow 0.67)$ and p_{T} $(0.12 \rightarrow 0.20)$ increase over time, with points distributed above the diagonal in both panels. This is the return map of a student whose orbit has not yet converged: the system is continuously revising its estimates upward as evidence of genuine learning accumulates interaction by interaction. A fixed label would misrepresent this student either as a weak learner (based on early interactions) or as a competent one (based on late interactions). The dynamic profile captures both states and the transition between them, enabling the tutorial model to progressively accelerate pacing and increase challenge level in real time—matched to the observed trajectory rather than to a predetermined category.

For *advancing learners*, the p_{L_0} return map shows early stabilisation: mastery is high from the first interactions (mean 0.74) and the orbit clusters around the diagonal with negligible net movement. The p_{T} panel declines moderately $(0.15 \rightarrow 0.11)$, indicating that the model assigns progressively smaller marginal learning rate estimates as opportunities for further gain diminish. The instructional implication is not derived from a presumed initial ability but from observed evidence: continued practice at the current difficulty level produces diminishing returns, and the tutorial model can route this student to more demanding content immediately, grounded in the trajectory rather than in a prior assignment.

These results underscore the utility of theory-grounded user modeling for ECTEL 2026's theme of Mindful TEL. By shaping learning technologies with intention, we ensure that deep learning's predictive power does not come at the cost of pedagogical transparency. The augmented user model provides the "diagnostic handles" required for educators to exercise agency within AI-mediated

Table 1: **Predictive performance (AUC) comparison.** Bold indicates best performance per dataset.

Model	AS2009	AS2015	AL2005	BDG2006	NIPS34
AKT	0.7825	0.7081	**0.8306**	**0.8208**	**0.8033**
ATKT	0.7715	**0.7810**	0.7995	0.7889	0.7665
DKT	0.7528	0.7031	0.8149	0.8015	0.7689
DKVMN	0.7440	0.7013	0.8054	0.7983	0.7673
SAKT	0.7263	0.6947	0.7880	0.7740	0.7517
SAINT	0.6921	0.6836	0.7775	0.7781	0.7873
gTransformer	**0.7831**	0.7078	0.8240	0.8148	0.8006

environments, transforming the "black box" into a collaborative cognitive engine that supports, rather than replaces, human expertise.

6 Experimental Setup

We evaluate gTransformer on five benchmark datasets with explicit knowledge component mappings: ASSISTments 2009 (AS2009, 4,217 students, 123 skills), ASSISTments 2015 (AS2015, 19,917 students, 100 skills), Algebra 2005 (AL2005, 574 students, 112 skills), Bridge to Algebra 2006 (BDG2006, 1,146 students, 493 skills), and NeurIPS 2020 Education Challenge (NIPS34, 4,918 students, 57 skills). We follow a standard five-fold cross-validation with an 80/20 train/test split, measuring performance via AUC using the question-level mean late-fusion protocol [9]. Baselines include DKT [11], DKVMN [13], SAKT [10], SAINT [2], AKT [4], and ATKT [5].

6.1 User Model: Theory-Grounded Knowledge Tracing

To quantify gTransformer's performance as a user model and the cost of pedagogical grounding, we define three metrics. Let p_{sup1} be ablated (non-grounded) supervised predictions, p_{sup2} grounded supervised predictions, p_{ref} interpretable BKT-based predictions, and p_{bkt} classical BKT predictions:

$$\mathcal{I}_{\text{drop}} = p_{\text{sup1}} - p_{\text{sup2}} \quad \text{(cost of grounding on neural capacity)} \tag{1}$$
$$\mathcal{I}_{\text{cost}} = p_{\text{sup2}} - p_{\text{ref}} \quad \text{(cost of switching to interpretable logic)} \tag{2}$$
$$\mathcal{I}_{\text{gain}} = p_{\text{ref}} - p_{\text{bkt}} \quad \text{(advantage over population-level BKT)} \tag{3}$$

Table 1 shows the comparative performance of gTransformer against baselines in the ablated (non-grounded) configuration. gTransformer achieves the best AUC on AS2009, ranks second on AL2005, BDG2006, and NIPS34, and third on AS2015, demonstrating that the architecture is competitive with the state of the art.

Table 2: **AUC–interpretability trade-offs.** $\mathcal{I}_{\text{drop}}$ = cost of grounding; $\mathcal{I}_{\text{cost}}$ = interpretability cost; $\mathcal{I}_{\text{gain}}$ = gain over BKT.

Dataset	p_{sup1}	p_{sup2}	p_{ref}	p_{bkt}	$\mathcal{I}_{\text{drop}}$	$\mathcal{I}_{\text{cost}}$	$\mathcal{I}_{\text{gain}}$
AS2009	0.7831	0.7814	0.7436	0.6097	0.2%	4.8%	22.0%
AS2015	0.7078	0.7070	0.6940	—	0.1%	1.8%	—
AL2005	0.8240	0.8237	0.7800	0.7215	0.0%	5.3%	8.1%
BDG2006	0.8148	0.8107	0.7810	0.6756	0.5%	3.6%	15.6%
NIPS34	0.8006	0.7987	0.7666	0.5729	0.2%	4.0%	33.8%
Mean	0.7861	0.7843	0.7530	0.6449	0.2%	3.9%	19.9%

Table 2 quantifies the trade-off. The interpretability drop ($\mathcal{I}_{\text{drop}}$) is negligible on average (0.2%), demonstrating that grounding constraints do not compromise the model's expressive capacity. The interpretability cost ($\mathcal{I}_{\text{cost}}$), i.e., the performance reduction when constraining predictions to BKT logic, averages 3.9% across datasets. Against population-level BKT, interpretable gTransformer predictions show a consistent advantage of 19.9% on average. These results establish that pedagogical interpretability can be achieved at a minimal performance cost.

6.2 Domain Model: Validating Theoretical Grounding

To verify that gTransformer's internal representations are genuinely grounded in the domain model (BKT), we implement a triple-validation methodology: (H2.1) Structural Alignment, assessed via diagnostic probing [6,1] measuring the linear recoverability of BKT constructs from hidden states; (H2.2) Semantic Alignment, assessed via Spearman's rank correlation (ρ) between grounded parameters and theoretical priors; and (H2.3) Functional Alignment, assessed via a composite confidence metric quantifying the reliability of interpretable predictions as substitutes for supervised ones. We validate using AS2009 as the reference dataset.

Structural Alignment (H2.1). Using diagnostic probing, we assess whether BKT constructs are linearly recoverable from final hidden states. Initial Mastery achieves fidelity $R^2 = 0.549$ ($r = 0.743$) with selectivity $\Delta R^2 = 0.619$, exceeding the 0.5 threshold that distinguishes genuine structural encoding from spurious correlations. Learn Rate achieves $R^2 = 0.515$ ($\Delta R^2 = 0.570$). These results confirm that BKT constructs are preserved as primary organising principles in the latent space.

Semantic Alignment (H2.2). Initial Mastery shows weaker monotonic alignment with theoretical priors ($\rho = 0.360$, MAE = 0.168), reflecting strong student-specific individualisation. Learn Rate demonstrates moderate alignment ($\rho = 0.544$, MAE = 0.092), preserving the pedagogical understanding of practice effects. Both parameters remain within pedagogical bounds, confirming that the model neither abandons nor mechanically reproduces theoretical priors.

Functional Alignment (H2.3). A composite confidence metric shows that 89.2% of student–skill pairs achieve medium or high confidence in interpretable predictions, indicating that theory-grounded outputs provide actionable diagnostic value.

6.3 Tutorial Model: Situation-Based Instruction

The theory-grounded parameters produced by gTransformer—student-specific estimates of initial mastery (p_{L_0}) and learn rate (p_{T})—provide the diagnostic handles through which an ITS tutorial model can implement *situation-based instruction*. Rather than applying a uniform instructional strategy, the tutorial model can route each student to a differentiated instructional pathway based on their learning situation as identified by gTransformer. The following section describes how this routing is operationalised through context-aware personalisation.

7 Context-Aware Personalisation

A key contribution of this work is demonstrating how gTransformer enables the ITS tutorial model to move from population-level strategies towards situation-based, context-aware instruction. Rather than assigning students to fixed ability groups, we identify four *learning situations* by partitioning the $(p_{\mathrm{L}_0}, p_{\mathrm{T}})$ parameter space along two dimensions—initial mastery and learning rate—on the AS2009 dataset: (1) *foundational learners* (low initial mastery, low learning rate), (2) *emerging learners* (low initial mastery, high learning rate), (3) *consolidating learners* (high initial mastery, low learning rate), and (4) *advancing learners* (high initial mastery, high learning rate). These learning situations are not static categories: because gTransformer continuously updates its parameter estimates as new interactions arrive, a student's assigned situation evolves with their trajectory. Each situation calls for a distinct tutorial strategy: foundational learners require targeted scaffolding; advancing learners are ready for accelerated pacing; intermediate situations benefit from adjusted challenge levels or motivational support.

Figure 4 shows mastery trajectories for students who provide identical response sequences to the same questions but belong to different learning situations. Classical BKT predictions (dotted grey lines) overlap perfectly, reflecting its Markovian constraint: given the same response sequence, BKT produces identical probability estimates regardless of learning history. gTransformer predictions (solid coloured lines) diverge based on the full interaction history, demonstrating the model's capacity to differentiate students not by what they answered, but by how they have learned across their entire trajectory.

This divergence has direct implications for the tutorial model. Each student interaction event can be routed to a different instructional pathway based on the learning situation assigned by gTransformer. A student whose trajectory places them in the foundational learning situation triggers a scaffolding strategy; a student identified as an advancing learner is routed towards accelerated content progression. Crucially, this routing is not static: as new interactions arrive, gTransformer updates its parameter estimates from the full longitudinal context, allowing the tutorial model to dynamically reassign students across situations as their learning trajectory evolves. This event-driven, context-aware routing is precisely what population-level models such as standard BKT cannot support—two students with an identical recent response pattern may receive different instructional interventions if their prior histories place them in different learning situations.

8 Discussion

8.1 Theory-Grounded Instructional Decisions

The results demonstrate that gTransformer addresses the accuracy–interpretability trade-off in KT by providing pedagogically grounded parameters at a minimal performance cost. This is directly relevant to the ECTEL 2026 theme of Mindful TEL: the model embodies the principle that learning technologies should be shaped with intention, ensuring that the cognitive engine of an ITS is anchored in established educational theory rather than opaque data-driven heuristics.

By operationalising BKT parameters (P_{L_0}, P_{T}) derived from deep learning context, gTransformer provides educators with the diagnostic handles through which an ITS can perform mindful instruction. Rather than acting as a black box that issues recommendations without justification, the system grounds its decisions in constructs that practitioners recognise and can validate. This aligns with broader EU imperatives for explainable, accountable AI in education [7], and responds directly to the concerns raised by the Artificial Intelligence Act's designation of educational AI as high-risk.

8.2 Beyond Static Labelling: Context-Aware Adaptation

A central contribution of this work is the shift from population-level labelling to situation-based diagnosis. The four learning situations—foundational, emerging, consolidating, and advancing—are not fixed assignments but dynamic summaries of each student's current longitudinal profile. The results presented in Section 7 demonstrate that gTransformer's context-aware predictions provide a higher-resolution understanding of the learning journey than population-level models. By encoding each student's longitudinal history into updatable parameters, the model acknowledges that students are dynamic learners whose instructional needs evolve continuously.

This enables educators to identify, for instance, students in the consolidating learning situation—high prior knowledge but low learning rate, which may signal disengagement or instructional mismatch—and adapt the pacing accordingly before significant learning gaps emerge. Conversely, students in the advancing situation can be identified for accelerated progression to prevent unnecessary remediation. Crucially, a student who responds positively to an intervention may transition from a foundational learning situation to an emerging one, and the tutorial model will update its routing accordingly. This intentional, responsive design addresses essential concerns about accountability and pedagogical integrity in automated instruction.

8.3 Pedagogical Integrity and Human Agency

The triple-validation methodology (RQ2) establishes that the model's internal representations genuinely internalise pedagogical theory rather than merely correlating with it. Structural alignment confirms BKT constructs as primary organising principles; semantic alignment confirms that grounded parameters retain their pedagogical semantics; and functional alignment quantifies the confidence with which educators can trust interpretable predictions. Together, these results support the claim that gTransformer is not merely a high-performance prediction system, but a theory-grounded diagnostic tool that can serve as the cognitive foundation of responsible, human-centred ITS.

9 Conclusions

This work demonstrates how gTransformer [8], a Grounded Transformer that bridges deep learning performance and pedagogical interpretability through *representational grounding*, can serve as the cognitive foundation of a theory-grounded ITS. We evaluated its capacity to instantiate the three core ITS components. As a user model, interpretable gTransformer predictions consistently surpass classical BKT with an average gain of 19.9% AUC, at a low interpretability cost of 3.9% relative to black-box architectures. As a domain model, gTransformer internalises BKT constructs as its primary organising principle in latent representations, confirmed by structural, semantic, and functional alignment. As a tutorial model foundation, gTransformer enables context-aware, situation-based instruction by capturing longitudinal learning patterns that Markovian models cannot represent.

As a foundation for TEL, gTransformer demonstrates that pedagogically justifiable student modeling is achievable without sacrificing the predictive power needed for effective adaptive instruction. The *representational grounding* methodology is domain-independent and may be extended to other theoretical frameworks beyond BKT or to other scientific fields where established theoretical models coexist with high-volume empirical data. Future research will investigate the integration of guess and slip parameters for finer-grained personalisation and the generalisation of gTransformer to alternative educational theories.

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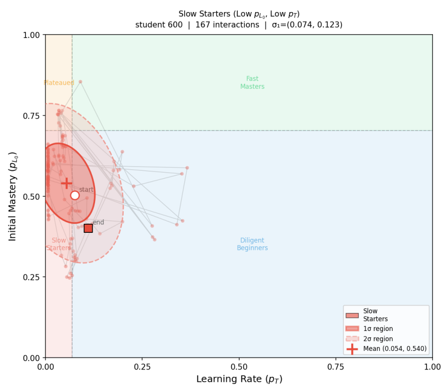
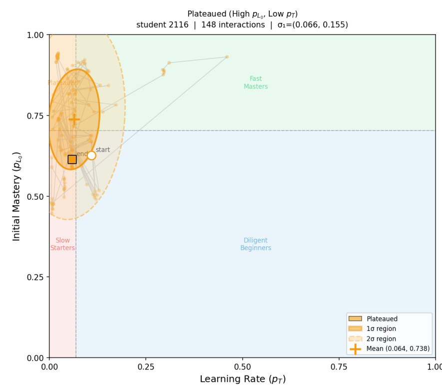
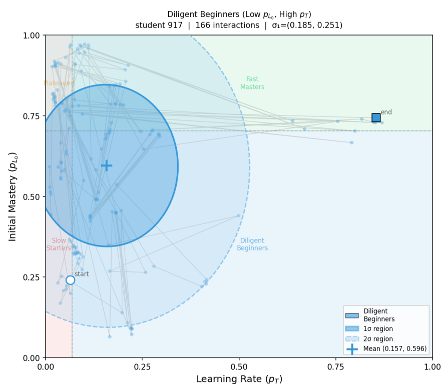


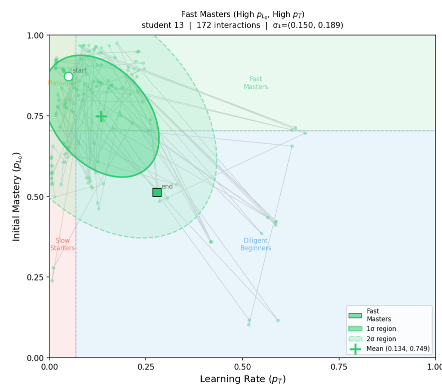
(a) Foundational



(b) Consolidating

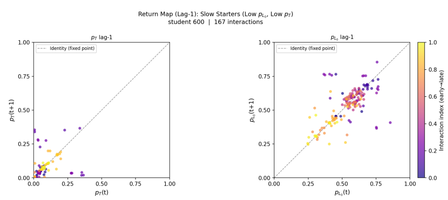


(c) Emerging

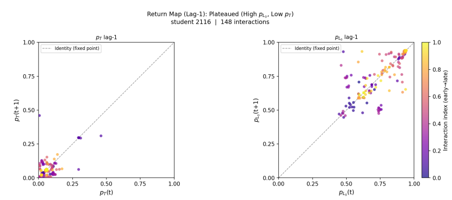


(d) Advancing

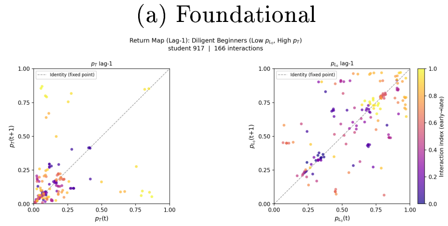
Fig. 2: **Parameter orbit ellipses.** Covariance ellipses (1σ and 2σ) for the representative student of each learning situation in $(p_{\mathrm{L}_0}, p_{\mathrm{T}})$ space. The centre marks the mean parameter position; ellipse shape encodes the extent and orientation of parameter variability across all interactions.



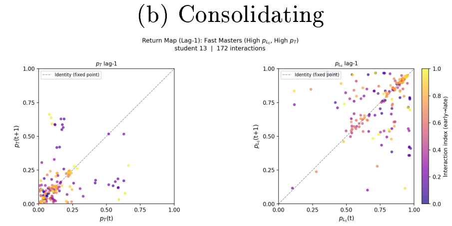
(a) Foundational



(b) Consolidating



(c) Emerging



(d) Advancing

Fig. 3: **Return maps.** Lag plots for the representative student of each learning situation. Each panel shows $p_{\mathrm{L}_0}(t+1)$ vs $p_{\mathrm{L}_0}(t)$ (left) and $p_{\mathrm{T}}(t+1)$ vs $p_{\mathrm{T}}(t)$ (right). Points are coloured from dark (early) to bright (late) interactions. The diagonal (dashed line) marks the fixed point; deviations above indicate increasing parameters and deviations below indicate decreasing ones.

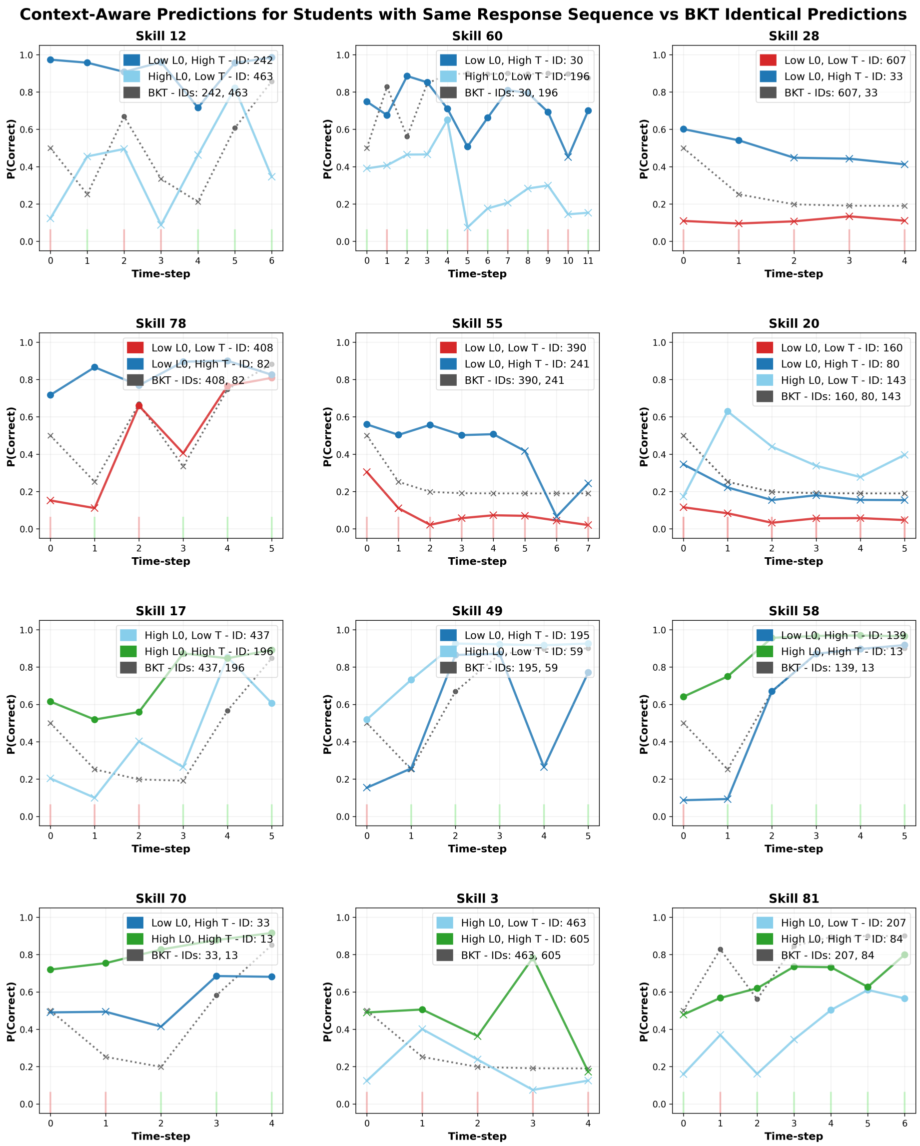


Fig. 4: **Context-aware prediction mosaic.** Each panel shows students with identical response sequences (green/red bars for correct/incorrect responses) but different learning situations. BKT predictions (dotted grey) overlap due to Markovian constraints; gTransformer predictions (solid coloured) diverge based on historical parameters (p_{L_0}, p_{T}). Circles indicate predicted correct responses; crosses indicate predicted failures.